SCHOLARSHIP ELIGIBILITY CHECKER

Python Program – Documentation

1. Problem Statement

Write a Python program to determine whether a student is eligible for a scholarship. The program accepts marks in five subjects, attendance percentage, and family income. A student is eligible only when every subject mark is at least 60, the overall percentage is at least 75%, attendance is at least 75%, and family income is not more than ■2,50,000. The program displays the eligibility result and the calculated percentage.

2. Assumptions

• Exactly five subject marks are entered.

• Marks are entered as numbers on a 0–100 scale.

• Attendance is entered as a percentage from 0 to 100.

• Family income is entered as a non-negative amount in rupees.

• The inputs are valid numeric values; input validation is not included in this program.

3. Algorithm

1. Start the program.

2. Create an empty list named marks.

3. Repeat five times: read a subject mark and append it to marks.

4. Read the attendance percentage and family income.

5. Calculate percentage = sum of marks / 5.

6. Check whether the minimum subject mark is at least 60, percentage is at least 75, attendance is at least 75, and income is at most 2,50,000.

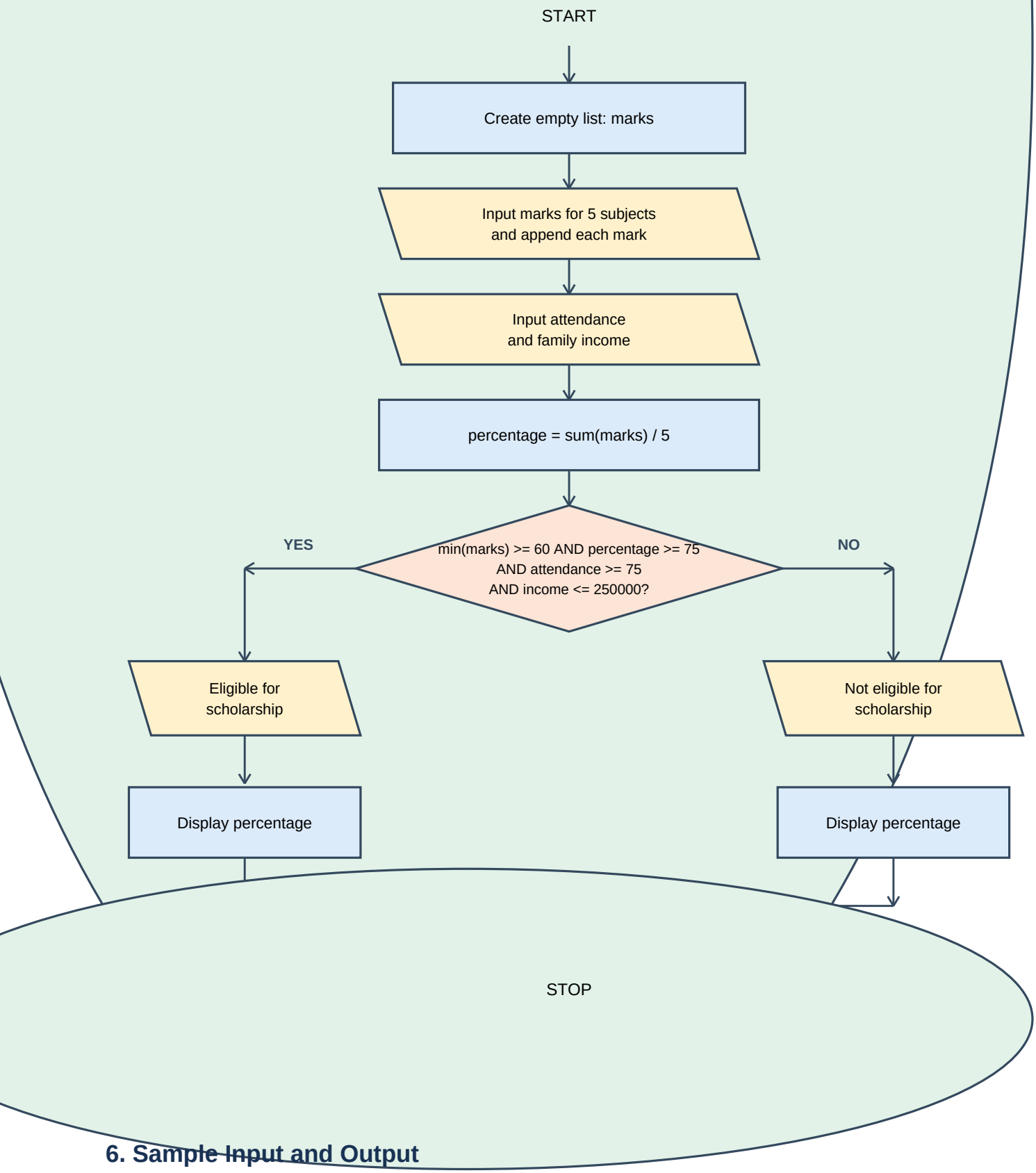
7. If all four conditions are true, display “Eligible for scholarship”; otherwise display “Not eligible for scholarship”.

8. Display the calculated percentage.

9. Stop.

4. Pseudocode

```
START
    marks ← empty list

    FOR i ← 1 TO 5
        INPUT m
        APPEND m TO marks
    END FOR

    INPUT attendance
    INPUT income

    percentage ← SUM(marks) / 5

    IF MIN(marks) >= 60 AND percentage >= 75
       AND attendance >= 75 AND income <= 250000 THEN
        PRINT "Eligible for scholarship"
    ELSE
        PRINT "Not eligible for scholarship"
    END IF

    PRINT "Percentage:", percentage
STOP
```



6. Sample Input and Output

Input	Sample value
Subject marks	80, 75, 90, 85, 70

Attendance	80
Family income	200000

```
Enter subject marks: 80
Enter subject marks: 75
Enter subject marks: 90
Enter subject marks: 85
Enter subject marks: 70
Enter attendance percentage: 80
Enter family income: 200000
Eligible for scholarship
Percentage: 80.0
```

7. Python Program

```
marks = []

for i in range(5):
    m = int(input("Enter subject marks: "))
    marks.append(m)

attendance = float(input("Enter attendance percentage: "))
income = float(input("Enter family income: "))

percentage = sum(marks) / 5

if min(marks) >= 60 and percentage >= 75 and attendance >= 75 and income <= 250000:
    print("Eligible for scholarship")
else:
    print("Not eligible for scholarship")

print("Percentage:", percentage)
```

8. Conclusion

The program checks the student's subject-wise marks, overall percentage, attendance, and family income against the given scholarship rules. It displays whether the student is eligible and prints the overall percentage. The sample data satisfies all conditions, so the output is “Eligible for scholarship”.